

An accurate triangular spectral element method-based numerical simulation for acoustic problems in complex geometries

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journals.sagepub.com/home/jae**Ximeng Ye** , **Guoliang Qin** and **Yazhou Wang**

Abstract

An accurate triangular spectral element method (TSEM) is developed to simulate acoustic problems in complex computational domains. With Fekete points and Koornwinder-Dubiner polynomials introduced, triangular elements are used in the present method to substitute quadrilateral elements in traditional spectral element method (SEM). The efficiency of discretizing complex geometry is enhanced while high accuracy of SEM is remained. The weak form of the second-order governing equations derived from the linearized Euler equations (LEEs) are solved, and perfectly matched layer (PML) boundary condition is implemented. Three benchmark problems with analytical solutions are employed to testify the exponential convergence rate, convenient implementation of solid wall boundary condition and capable discretization in complex geometries of the present method respectively. An application on Helmholtz resonator (HR) is presented as well to demonstrate the possibility of using the present method in practical engineering. The numerical resonance frequency of HR reaches an excellent agreement with the theoretical result.

Keywords

Triangular spectral element method, acoustic wave propagation, high accuracy, perfectly matched layer, Helmholtz resonator

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Introduction

Computational aeroacoustics (CAA), which raises public concern in recent years, is a potential tool for practical engineering that ranges from prediction and reduction of jet noise, propeller noise and combustion noise to investigations on bionic silent flight. Two representative challenges that CAA faces are: (1) the small magnitude of sound field demands the high-order numerical method; (2) the complexity of computational domain in practical applications requires a flexible spatial discretization strategy. Considerable efforts have been made over these issues.

Most CAA methods inherit from computational fluid dynamics (CFD) methods, but higher accuracy, lower dispersion and lower dissipation of numerical methods are expected. This is because the acoustic fluctuation is around three or four orders of magnitude smaller than the flow fluctuation.¹ The most commonly used method may be high-order finite difference methods (FDMs), which are represented by the dispersion-relation-preserving (DRP) scheme proposed by Tam and Webb.² Hu³ studied the Couette flow and roll-up vortices using DRP. Similar scheme can also be found in study about the sound scattering around circular cylinders,⁴ and study focusing on time-reversal simulation of a reflecting wall.⁵ A high-order FDM with spectral-like resolution is reported by Bai et al.⁶ with benchmark problems presented. Application of high-order finite difference schemes can also be found in acoustic source localization with time-reversal and beamforming method.^{7,8} However, one limitation of FDM is that schemes with different orders are constructed independently with each other. Therefore, increasing the order of FDM in calculation requires additional efforts to construct corresponding high-order scheme, especially corresponding schemes for points near the boundary,⁴ which may lead to inefficiency when pursuing higher accuracy in CAA problems. In recent years, the spectral element method (SEM), a combination of finite element method (FEM) and spectral method, overcomes the drawback of FDM thanks to its characteristic exponential accuracy. That is to say, the calculation error decreases exponentially as the interpolation order goes up. This feature makes SEM more accurate than FEM (which will be discussed later) and more efficient and convenient than FDM. There are some reports about successful implementation of SEM in CFD^{9,10} and CAA^{11,12} problems where high accuracy is required.

Most CAA problems are performed in complex computational domains. Unfortunately, numerical methods, including SEM, are mainly carried out with structured grids, which are difficult to discretize irregular regions. The traditional way of enhancing the discretization capability is to divide the computational domain into several blocks with special topology design, which leads to large time consumption in mesh generation. Therefore, methods with generality while dealing with different geometries are needed. There are generally three solutions. The first solution is to implement special boundary treatment on structured grids represented by immersed boundary method (IBM),^{13,14} where boundaries are not required to be at the mesh node. The second approach is to change the form of meshes, involving using triangular elements¹⁵ or curved elements and using 1 D boundary elements (in boundary element method^{16,17}). The third way is meshless methods including smoothed particle hydrodynamics (SPH)¹⁸ and related improved methods, which use particles instead of meshes to describe geometries and hence keep away from the root of the problem: mesh generation. Among all above solutions, the best choice to enhance the spatial discretizing capability of SEM may be the second one, especially the triangular elements or the curved elements, which are both simple and convenient compared with other solutions. In this way,

SEM becomes an effective tool for CAA problems, with both high accuracy of traditional SEM and capable spatial discretization of different elements. In our previous work,¹⁹ SEM with curved quadrilateral elements was carried out for wave propagation with an absorbing boundary condition (ABC). However, obvious distortion of wave can be observed near the artificial boundaries due to the accuracy inefficiency of ABC implemented. In the present paper, we focus on triangular elements, i.e. triangular spectral element method (TSEM). Perfectly matched layer (PML), another type of non-reflective boundary condition which is theoretically reflectionless for waves of any angle and frequency,³ is chosen to enhance accuracy near the artificial boundary.

There are generally two ways to implement triangular elements in SEM. The first way is to map quadrature points such as Gauss-Lobatto points or Gauss-Chebyshev points which originally distribute in quadrilateral elements into triangles.²⁰ The main drawback of this approach is, distribution of points is highly uneven, which leads to nonuniformity of accuracy inside the triangle element. In this paper, we mainly focus on another approach. This approach employs orthogonal polynomials and interpolation points established in triangles instead of quadrangles to guarantee the uniformity of point distribution. In 1991, Dubiner²¹ first proposed such a set of polynomials based on Jacobi polynomials, named Koornwinder-Dubiner (KD) polynomials. The most commonly used interpolation points are Fekete points. Fekete points are proved to be Gauss-Lobatto-Legendre points in quadrilaterals,²² which permits the implementation of hybrid elements.

In CAA, generally the first step is to simulate the flow field using Navier-Stokes equation, then calculate the sound source and perform acoustic field simulation. So far, there are a few numerical investigations on solving fluid flow problems^{23–25} using TSEM, which proved effectiveness of TSEM in flow field simulation. In the present paper, we mainly focus on the second step of acoustic wave calculation. Recently, a relevant work is reported by Pind et al.²⁶ about room acoustics using TSEM. Nevertheless, boundary conditions for aeroacoustics are quite different from room acoustics. In the field of CAA, it's rare to find application of TSEM as far as we know.

In this paper, an accurate triangular spectral element method to solve acoustic problems in complex domains is presented. The aim of the present paper is to develop TSEM with both high accuracy and capable spatial discretization of complex geometries to handle the two challenges in CAA. The unique exponential convergence characteristic of SEM guarantees high accuracy, while the use of triangular elements provides powerful capability of discretizing complex geometry. Benchmark problems with analytical solutions are performed as validation, and an application of acoustic wave propagation in Helmholtz resonator is also presented. It is believed TSEM will be helpful on the way addressing the two challenges of accuracy and spatial discretization in acoustic wave simulation.

Physical problem

Governing equations

The fluid is assumed to be inviscid, and the mass force is neglected. The flow variables can be expressed as:

$$\mathbf{U} = (\rho u v p)^T = \mathbf{U}_0 + \mathbf{U}' \quad (1)$$

where the subscript ‘0’ represents variables of mean flow, and the superscript ‘’ denotes variables of perturbation caused by sound propagation. The perturbed variables are supposed to be much smaller than mean flow ones with the theory of small disturbance. Based on above assumptions, the full 2D Navier-Stokes equations can be simplified into linearized Euler equations (LEEs) expressed as:

$$\frac{\partial \mathbf{U}'}{\partial t} + \frac{\partial \mathbf{E}'}{\partial x} + \frac{\partial \mathbf{F}'}{\partial y} = \mathbf{S} \quad (2)$$

$$\mathbf{E}' = \begin{pmatrix} Ma_x \rho' + u' \\ Ma_x u' + p' \\ Ma_x v' \\ Ma_x p' + u' \end{pmatrix}, \quad \mathbf{F}' = \begin{pmatrix} Ma_y \rho' + v' \\ Ma_y u' \\ Ma_y v' + p' \\ Ma_y p' + v' \end{pmatrix}, \quad \mathbf{S} = \begin{pmatrix} S_1 \\ S_2 \\ S_3 \\ S_4 \end{pmatrix} \quad (3)$$

where Ma_x and Ma_y stand for the Mach numbers of the mean flow in x and y directions respectively, and $\mathbf{S}$ is the source term which depends on problems. To simplify the formulations, the superscript ‘’ will be omitted in the following discussion. All the variables in equation (2) are nondimensionalized by following factors: $\rho \sim \rho_\infty$, $(u, v) \sim c_\infty$, $p \sim \rho_\infty c_\infty^2$, $(x, y) \sim L$, $t \sim L/c_\infty$, where ρ_∞ is density of the mean flow, c_∞ is the sound speed, and L is a characteristic length that depends on the problem.

To focus only on the perturbed pressure p , we eliminate u and v in equation (2) and get:

$$\frac{\partial^2 p}{\partial t^2} + 2Ma_x \frac{\partial^2 p}{\partial t \partial x} + \left[(Ma^2 - 1) \frac{\partial^2 p}{\partial x^2} - \frac{\partial^2 p}{\partial y^2} \right] = S \quad (4)$$

with

$$S = \frac{\partial S_2}{\partial x} + \frac{\partial S_3}{\partial y} + Ma \frac{\partial S_4}{\partial x} + \frac{\partial S_4}{\partial t} \quad (5)$$

In equation (4), for the sake of convenience, we assume a mean flow with speed along the x direction in this paper, that is to say, $Ma_y = 0$ and $Ma_x = Ma$. For situations with $Ma_y \neq 0$, the coordinate system will be rotated till the mean flow is along x direction. The complete derivation of equation (4) can be found in Appendix 1.

One of the advantages of using the second-order governing equations equation (4) is the reduction of quantities from three (p, u, v) to one (p only) compared with LEEs. If values of u and v are needed, they can be obtained after calculation of p . In this way, less computational resource and less time consumption are required. Another benefit of (equation (4)) is the convenience of solid wall boundary condition implementation, which will be discussed later.

Boundary conditions

PML boundary condition. The perfectly matched layer (PML) boundary condition (B.C.) was first proposed by Berenger²⁷ in 1994 for electromagnetic problems. In recent years, PML has

been extended to applications in wider range^{28–30} including CAA, and considerable improvements were made to promote the computational efficiency.^{31,32} In this paper, an improved split-PML developed by Komatsitsch and Tromp³³ is applied. In Ref.³³, this method was originally developed for seismic wave simulation with second-order governing equation. In this section, a similar idea is used to derive the PML-included governing equation for acoustic problems based on equation (4).

The basic idea of PML is to introduce an ordinate transformation, here we take the 1D situation for example:

$$x = x' - \frac{i}{\omega} \int_{x_0}^{x'} \sigma_x dx' \quad (6)$$

where ω is the angular velocity in the frequency domain, σ_x is called the absorption coefficient, i is the imaginary unit, and x_0 is the starting point of a wave in the form: $p = \exp[i(\omega t - kx)]$.

Substitute equation (6) into the expression of the wave:

$$p = \exp \left[i(\omega t - kx') - \frac{k}{\omega} \int_{x_0}^{x'} \sigma_x dx' \right] \quad (7)$$

The ordinate transformation is equivalent to reducing the amplitude of the wave when $\sigma_x > 0$. This is where PML comes from. See Refs.^{3,33} for more information.

Substitute equation (6) into the derivative with respect to x , we have:

$$\frac{\partial}{\partial x} = \frac{i\omega}{i\omega + \sigma_x} \frac{\partial}{\partial x'} \quad (8)$$

Formula in y direction could be obtained in the similar way:

$$\frac{\partial}{\partial y} = \frac{i\omega}{i\omega + \sigma_y} \frac{\partial}{\partial y'} \quad (9)$$

Combined with equations (8) and (9), equation (6) is rewritten as:

$$\frac{\partial^2 p_1}{\partial t^2} + 2\sigma_x \frac{\partial p_1}{\partial t} + 2Ma \frac{\partial^2 p}{\partial t \partial x} + \sigma_x^2 p_1 + 2Ma\sigma_x \frac{\partial p}{\partial x} - (1 - Ma^2) \frac{\partial^2 p}{\partial x^2} = (Ma^2 - 1)f + \frac{S}{2} \quad (10)$$

$$\frac{\partial^2 p_2}{\partial t^2} + 2\sigma_y \frac{\partial p_2}{\partial t} + \sigma_y^2 p_2 - \frac{\partial^2 p}{\partial y^2} = -g + \frac{S}{2} \quad (11)$$

$$\frac{\partial f}{\partial t} + \sigma_x f = \frac{\partial \sigma_x}{\partial x} \frac{\partial p}{\partial x} \quad (12)$$

$$\frac{\partial g}{\partial t} + \sigma_y g = \frac{\partial \sigma_y}{\partial y} \frac{\partial p}{\partial y} \quad (13)$$

where p_1 and p_2 are components of p , hence $p = p_1 + p_2$. f and g are introduced intermediate variables. σ_x and σ_y are absorption coefficients in x and y directions, which equal zero in the interior computational region, and are positive in PML. The derivation of equations (10) to (13) is presented in Appendix 2.

The PML-included governing equations consist of equations (10) to (13) and four independent unknown quantities (p_1, p_2, f, g) , so it's solvable. The value of p will be calculated by $p = p_1 + p_2$ after p_1 and p_2 are solved.

Solid wall boundary condition. To find the solid wall B.C. for the second-order governing equations, let's recall the impermeable wall boundary for inviscid flow in the original form of governing equations (equation (2)):

$$(u, v) \cdot \mathbf{n} = 0 \quad (14)$$

where $\mathbf{n}$ is the external normal direction of solid wall.

After derivation, the corresponding solid wall B.C. for the second-order governing equations is:

$$\frac{\partial p}{\partial n} = 0 \quad (15)$$

Numerical methodology

To implement TSEM, orthogonal polynomials and the corresponding interpolation points on the standard triangle are required. A standard triangle element Ω is defined as $\Omega = \{(x, y) | -1 \leq x, y \leq 1, x + y \leq 0\}$, as showed in Figure 1.

KD polynomials and Fekete points. The KD polynomials can be written as:

$$\phi_m(x, y) = c_{ij} P_i^{0,0}(\bar{x}) P_j^{2i+1,0} \left(\frac{1-y}{2} \right)^i, \quad (x, y) \in \Omega \quad (16)$$

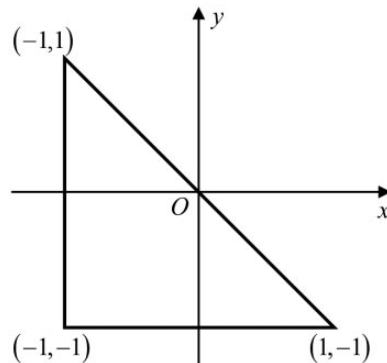


Figure 1. A standard triangle element for TSEM.

where m is a function of i and j , $0 \leq i + j \leq N$, N is known as the order of KD polynomials. $P_n^{\alpha, \beta}(x)$ is Jacobi polynomial in n th order, $\bar{x}$, $\bar{y}$ and c_{ij} are given by:

$$\bar{x} = \frac{2x + y + 1}{1 - y}, \quad \lim_{x \rightarrow -1, y \rightarrow 1} \bar{x} = -1, \quad \bar{y} = y, \quad c_{ij} = \sqrt{\frac{(2i+1)(i+j+1)}{2}} \quad (17)$$

It's obvious when the order is N , the number of KD polynomials $Np = (N+1)(N+2)/2$. KD polynomials are orthonormal in Ω with the weight $\omega = 1$:

$$\int_{\Omega} \phi_m(x, y) \phi_n(x, y) dS = \delta_{mn} \quad (18)$$

The Lagrange basis of these points is $L_i(x, y)$ which satisfies $L_i(x, y) \leq 1$ in the standard triangle, and $L_i(x, y) = 1$ is only established when $(x, y) = (x_i, y_i)$, where $(x_i, y_i) (i = 1, 2, \dots, Np)$ are a set of points in Ω . KD polynomials can be presented by (x_i, y_i) and L_i :

$$\phi_i(x, y) = \sum_{j=1}^{Np} \phi_i(x_j, y_j) L_j(x, y) \quad (19)$$

The matrix form is:

$$\Phi = \mathbf{V}^T \mathbf{L} \quad (20)$$

where $\mathbf{V}$ is called the Vandermonde matrix, $V_{ij} = \phi_j(x_i, y_i)$, the superscript 'T' denotes transposition.

Note that the form of Lagrange basis is unknown. Using equation (20), the Lagrange basis can be evaluated with Cramer's Rule:

$$L_j = \frac{|\mathbf{V}_j^T|}{|\mathbf{V}^T|} \quad (21)$$

where $\mathbf{V}_j^T$ is a matrix gained from replacing the j th row of $\mathbf{V}^T$ is by Φ .

To make sure L_j meets the definition of Lagrange basis that $L_j(x, y) \leq 1$ in the standard triangle, and $L_j(x, y) = 1$ is only established when $(x, y) = (x_i, y_i)$, the maximum value of $|\mathbf{V}_j^T|$ should be reached at the nodal points (x_i, y_i) . In this way, (x_i, y_i) are carried out by solving an optimization problem: find a set of points $(x_i, y_i) (i = 1, 2, \dots, Np)$ to make $|\mathbf{V}_j^T|$ maximize. This set of points are called Fekete points.

The analytical solution of Fekete points is hardly known. In practice, Fekete points are generally computed numerically.^{22,34} In this paper, Fekete points calculated by Briani et al.³⁵ are used.

With KD polynomials and Fekete points, any continuous function f defined on Ω can be approximated:

$$f(x, y) = \sum_{i=1}^{Np} f^{(i)} L_i(x, y) = \sum_{i=1}^{Np} \sum_{j=1}^{Np} f^{(i)} (V^{-1})_{ji} \phi_j(x, y) \quad (22)$$

where $f^{(i)} = f(x_i, y_i)$. For non-standard elements, we map them into the standard element before the above procedures.

Spatial discretization. The weak form of governing equations (equations (10) to (13)) in Galerkin approach is: to find trial functions $(p_1, p_2, f, g) \in H^1(\Omega) \times H^1(\Omega) \times H^1(\Omega) \times H^1(\Omega)$, such that

$$\begin{aligned} & (1 - Ma^2) \int_{\Omega} \frac{\partial p}{\partial x} \frac{\partial v}{\partial x} d\Omega + \int_{\Omega} \left(\frac{\partial^2 p_1}{\partial t^2} + 2\sigma_x \frac{\partial p_1}{\partial t} + 2Ma \frac{\partial^2 p}{\partial x \partial t} + \sigma_x^2 p_1 + 2Ma\sigma_x \frac{\partial p}{\partial x} \right) v d\Omega \\ & = \int_{\Omega} \left[(Ma^2 - 1)f + \frac{S}{2} \right] v d\Omega + (1 - Ma^2) \int_{\Gamma_2} \frac{\partial p}{\partial x} n_x v ds \end{aligned} \quad (23)$$

$$\int_{\Omega} \frac{\partial p}{\partial y} \frac{\partial v}{\partial y} d\Omega + \int_{\Omega} \left(\frac{\partial^2 p_2}{\partial t^2} + 2\sigma_y \frac{\partial p_2}{\partial t} + \sigma_y^2 p_2 \right) v d\Omega = \int_{\Omega} \left(-g + \frac{S}{2} \right) v d\Omega + \int_{\Gamma_2} \frac{\partial p}{\partial y} n_y v ds \quad (24)$$

$$\int_{\Omega} \left(\frac{\partial f}{\partial t} + \sigma_x f \right) v d\Omega = \int_{\Omega} \frac{\partial \sigma_x}{\partial x} \frac{\partial p}{\partial x} v d\Omega \quad (25)$$

$$\int_{\Omega} \left(\frac{\partial g}{\partial t} + \sigma_y g \right) v d\Omega = \int_{\Omega} \frac{\partial \sigma_y}{\partial y} \frac{\partial p}{\partial y} v d\Omega \quad (26)$$

where $H^1(\Omega)$ is the Sobolev space on Ω , Γ_2 is Neumann boundary, i.e., the solid wall boundary in the problem we concern; n_x and n_y are projections of the unit normal vector $\mathbf{n}$ of solid wall; v is the test function which is set as Lagrange basis in Galerkin approach:

$$v = L_i(x, y), \quad i = 1, 2, \dots, Np \quad (27)$$

To implement solid wall B.C $\partial p / \partial n = 0$ automatically in the governing equation, we need to deal with terms concerned with Neumann boundaries in equations (23) and (24) ($\int_{\Gamma_2} \partial p / \partial x \cdot n_x v ds$ and $\int_{\Gamma_2} \partial p / \partial y \cdot n_y v ds$). When $Ma=0$, sum the term $\int_{\Gamma_2} \partial p / \partial x \cdot n_x v ds$ and $\int_{\Gamma_2} \partial p / \partial y \cdot n_y v ds$, then we have:

$$\int_{\Gamma_2} \left(\frac{\partial p}{\partial x} n_x + \frac{\partial p}{\partial y} n_y \right) v ds = \int_{\Gamma_2} \frac{\partial p}{\partial n} v ds = 0 \quad (28)$$

Note that in split-PML, we only need to make sure the sum of the two split equations equals to the original governing equation, so it's possible to move $\int_{\Gamma_2} \partial p / \partial x \cdot n_x v ds$ into equation (24) without changing the summation of equations (23) and (24). When we sum equations (23) and (24) up for the interior region ($\sigma_x, \sigma_y = 0$ and $f, g = 0$), the governing equation is:

$$\int_{\Omega} \frac{\partial p}{\partial x} \frac{\partial v}{\partial x} d\Omega + \int_{\Omega} \frac{\partial p}{\partial y} \frac{\partial v}{\partial y} d\Omega + \int_{\Omega} \frac{\partial^2 p}{\partial t^2} v d\Omega = \int_{\Omega} S v d\Omega + \int_{\Gamma_2} \frac{\partial p}{\partial n} n v ds \quad (29)$$

which is the same as the weak form of equation (4) when $Ma=0$. So the movement of $\int_{\Gamma_2} \partial p / \partial x \cdot n_x v ds$ into equation (24) is available. In this way, the value of p_1 and p_2 may be changed a little bit, but value of p in the interior region is not effected. The advantage of this operation is: the term of the Neumann boundary in equation (24) becomes $\int_{\Gamma_2} \partial p / \partial n \cdot v ds = 0$. Therefore, to implement the solid wall B.C., we only need to ignore the boundary integral terms. That is to say, a theoretically exact solid wall B.C. could be brought in TSEM with no treatment.

In addition to the situation when $Ma=0$, there are several situations where solid wall B.C. can also be treated easily. For example, when the solid wall is parallel to x -axis or y -axis, the term $\int_{\Gamma_2} \partial p / \partial y \cdot n_y v ds = \int_{\Gamma_2} \partial p / \partial n \cdot v ds = 0$. Such simplification can be done for all the circumstances when the terms about Neumann boundary in equations (23) and (24) can be transformed into some kinds of format related to $\partial p / \partial n$. Such convenient implementation of solid wall B.C. is the reason for using the second order governing equation equation (4) of p instead of LEEs (equation (2)). On the contrary, additional schemes or other treatment are needed to maintain the accuracy near the solid wall in the most commonly used FDM.⁴ Therefore, TSEM is more convenient than FDM from the point of view of solid wall B.C. treatment.

Replace quantities in equations (23) – (26) with their spectral approximations (equation (22)), governing equations with spatial discretized in Ω are:

$$[(1 - Ma^2)\mathbf{A}_e \mathbf{p} + 2Ma\mathbf{D}_e(\dot{\mathbf{p}} + \sigma_x \mathbf{p}) + \mathbf{F}_e(\ddot{\mathbf{p}}_1 + 2\sigma_x \dot{\mathbf{p}}_1 + \sigma_x^2 \mathbf{p}_1)] = (Ma^2 - 1)\mathbf{F}_e \mathbf{f} + \frac{\mathbf{S}_e}{2} \quad (30)$$

$$\mathbf{B}_e \mathbf{p} + \mathbf{F}_e(\ddot{\mathbf{p}}_2 + 2\sigma_y \dot{\mathbf{p}}_2 + \sigma_y^2 \mathbf{p}_2) = -\mathbf{F}_e \mathbf{g} + \frac{\mathbf{S}_e}{2} \quad (31)$$

$$\mathbf{F}_e(\dot{\mathbf{f}} + \sigma_x \mathbf{f}) = \mathbf{D}_e \frac{\partial \sigma_x}{\partial x} \mathbf{p} \quad (32)$$

$$\mathbf{F}_e(\dot{\mathbf{g}} + \sigma_y \mathbf{g}) = \mathbf{E}_e \frac{\partial \sigma_y}{\partial y} \mathbf{p} \quad (33)$$

with

$$A_{e,ij} = \sum_{m=1}^{Np} \sum_{n=1}^{Np} V_{mi}^{-1} V_{nj}^{-1} \int_{\Omega} \frac{\partial \phi_m}{\partial x} \frac{\partial \phi_n}{\partial x} d\Omega \quad (34)$$

$$B_{e,ij} = \sum_{m=1}^{Np} \sum_{n=1}^{Np} V_{mi}^{-1} V_{nj}^{-1} \int_{\Omega} \frac{\partial \phi_m}{\partial y} \frac{\partial \phi_n}{\partial y} d\Omega \quad (35)$$

$$D_{e,ij} = \sum_{m=1}^{Np} \sum_{n=1}^{Np} V_{mi}^{-1} V_{nj}^{-1} \int_{\Omega} \phi_m \frac{\partial \phi_n}{\partial x} d\Omega \quad (36)$$

$$E_{e,ij} = \sum_{m=1}^{Np} \sum_{n=1}^{Np} V_{mi}^{-1} V_{nj}^{-1} \int_{\Omega} \phi_m \frac{\partial \phi_n}{\partial y} d\Omega \quad (37)$$

$$F_{e,ij} = \sum_{m=1}^{Np} \sum_{n=1}^{Np} V_{mi}^{-1} V_{nj}^{-1} \int_{\Omega} \phi_m \phi_n d\Omega \quad (38)$$

$$\mathbf{S}_e = \int_{\Omega} \mathbf{S} \mathbf{L} d\Omega \quad (39)$$

where bold quantities represent vectors formed by their value on interpolation points, for example, $\mathbf{p} = (p^{(1)}, p^{(2)}, \dots, p^{(Np)})^T$. The subscript 'e' denotes that these matrices are calculated in only one element.

To take contributes of all the elements into consideration, we assemble $Np \times Np$ matrices $\mathbf{A}_e - \mathbf{F}_e$ following the assembly law in FEM into $Nd \times Nd$ matrices $\mathbf{A} - \mathbf{F}$, where Nd is the total number of interpolation points. Similarly, $\mathbf{S}$ is assembled by $\mathbf{S}_e$.

Temporal scheme. The implicit Newmark method is employed in this paper. Focusing on time derivative, equations (30) and (31) can be expressed as:

$$\mathbf{M}\ddot{\mathbf{q}} + \mathbf{C}\dot{\mathbf{q}} + \mathbf{K}\mathbf{q} = \mathbf{Q} \quad (40)$$

with

$$\mathbf{M} = \begin{pmatrix} \mathbf{F} & 0 \\ 0 & \mathbf{F} \end{pmatrix} \quad (41)$$

$$\mathbf{C} = \begin{pmatrix} 2Ma\mathbf{D} + 2\sigma_x\mathbf{F} & 2Ma\mathbf{D} \\ 0 & 2\sigma_y\mathbf{F} \end{pmatrix} \quad (42)$$

$$\mathbf{K} = \begin{pmatrix} (1 - Ma^2)\mathbf{A} + 2Ma\sigma_x\mathbf{D} + \sigma_x^2\mathbf{F} & (1 - Ma^2)\mathbf{A} + 2Ma\sigma_x\mathbf{D} \\ \mathbf{B} & \mathbf{B} + \sigma_y^2\mathbf{F} \end{pmatrix} \quad (43)$$

$$\mathbf{Q} = \begin{pmatrix} (Ma^2 - 1)\mathbf{F}\mathbf{f} + \frac{\mathbf{S}}{2} \\ \mathbf{F}\mathbf{g} + \frac{\mathbf{S}}{2} \end{pmatrix} \quad (44)$$

where $\mathbf{q} = (p_1^{(1)}, p_1^{(2)}, \dots, p_1^{(Nd)}, p_2^{(1)}, p_2^{(2)}, \dots, p_2^{(Nd)})^T$, $\dot{\mathbf{q}}$ and $\ddot{\mathbf{q}}$ are the first and second order partial derivatives of time.

The implicit Newmark scheme is expressed as follows:

$$(c_0\mathbf{M} + c_1\mathbf{C} + \mathbf{K})\mathbf{q}^{t+\Delta t} = \mathbf{Q}^{t+\Delta t} + \mathbf{M}(c_0\mathbf{q}^t + c_2\dot{\mathbf{q}}^t + c_3\ddot{\mathbf{q}}^t) + \mathbf{C}(c_1\mathbf{q}^t + c_4\dot{\mathbf{q}}^t + c_5\ddot{\mathbf{q}}^t) \quad (45)$$

$$\dot{\mathbf{q}}^{t+\Delta t} = c_0(\mathbf{q}^{t+\Delta t} - \mathbf{q}^t) - c_2\ddot{\mathbf{q}}^t - c_3\ddot{\mathbf{q}}^t \quad (46)$$

$$\dot{\mathbf{q}}^{t+\Delta t} = \dot{\mathbf{q}}^t + c_6\ddot{\mathbf{q}}^t + c_7\ddot{\mathbf{q}}^{t+\Delta t} \quad (47)$$

where $c_0 = \frac{1}{\alpha\Delta t^2}$, $c_1 = \frac{\delta}{\alpha\Delta t}$, $c_2 = \frac{1}{\alpha\Delta t}$, $c_3 = \frac{1}{2\alpha} - 1$, $c_4 = \frac{\delta}{\alpha} - 1$, $c_5 = \left(\frac{\delta}{2\alpha} - 1\right)\Delta t$, $c_6 = (1 - \delta)\Delta t$, $c_7 = \delta\Delta t$. α and δ are constants controlling the accuracy and stability of the scheme. In this paper, we set $\alpha = 0.49$, $\delta = 0.9$ which guarantees the absolute stability of calculation.³⁶

$\mathbf{q}^0$ and $\dot{\mathbf{q}}^0$ will be given as the initial condition, and thus $\ddot{\mathbf{q}}^0$ can be calculated by:

$$\mathbf{M}\ddot{\mathbf{q}}^0 = (\mathbf{Q}^0 - \mathbf{K}\mathbf{q}^0 - \mathbf{C}\dot{\mathbf{q}}^0) \quad (48)$$

The additional equations (25) and (26) with only first-order time derivation is dealt with the second-order explicit Adams-Bashford scheme:

$$\mathbf{f}^{t+\Delta t} = \mathbf{f}^t + \Delta\mathbf{f}^t \quad (49)$$

$$\mathbf{g}^{t+\Delta t} = \mathbf{g}^t + \Delta\mathbf{g}^t \quad (50)$$

with

$$\Delta\mathbf{f}^t = \frac{\Delta t}{2} \left[\mathbf{F}^{-1}\mathbf{D} \frac{\partial\sigma_x}{\partial x} (3\mathbf{p}^t - \mathbf{p}^{t-\Delta t}) - \sigma_x(3\mathbf{f}^t - \mathbf{f}^{t-\Delta t}) \right] \quad (51)$$

$$\Delta\mathbf{g}^t = \frac{\Delta t}{2} \left[\mathbf{F}^{-1}\mathbf{E} \frac{\partial\sigma_y}{\partial y} (3\mathbf{p}^t - \mathbf{p}^{t-\Delta t}) - \sigma_y(3\mathbf{g}^t - \mathbf{g}^{t-\Delta t}) \right] \quad (52)$$

The whole process of solving acoustic wave propagation problems using the present TSEM is given as a flowchart in Figure 2.

Numerical examples and discussions

In this section, several numerical examples are presented. The first three examples are benchmark problems with analytical solutions. They will help to testify features of the present TSEM including: a) high accuracy and exponential convergence, b) the convenient solid wall B.C. implementation and c) capability of complex geometry discretization. Finally, we perform an practical application of TSEM on noise reduction of Helmholtz resonators to demonstrate the possibility of application prospect of this present TSEM.

Sound propagation in an infinite plane

The general purpose of the first example is to demonstrate the numerical accuracy of TSEM and validate the effectiveness of PML combining with TSEM. An initial perturbation in an

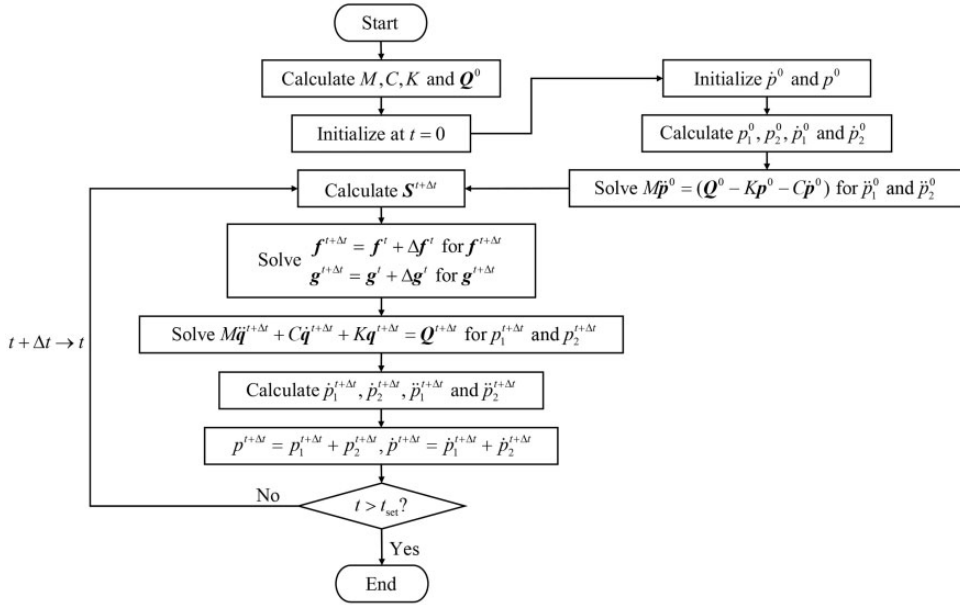


Figure 2. Flowchart of TSEM for acoustic wave propagation with Newmark scheme in time.

infinite plane is described as:

$$p(x, y, 0) = \exp \left[-\frac{(x - x_s)^2 + (y - y_s)^2}{b^2} \right] \quad (53)$$

where $(x_s, y_s) = (0, 0)$, $b^2 = 4/\ln 2$.

The governing equations are solved with $Ma = 0$, $S = 0$. The absorption coefficient σ_x is set as $\sigma_x = 10(x - x_0)^2/\delta^2$, where x_0 is the coordinate of the interface of PML and interior region (see Figure 3), δ is the thickness of PML, $\delta = 2$. σ_y is defined in the similar way. The interior computational domain is $[0, 10] \times [0, 10]$, where only one quarter of the total domain is calculated considering the symmetry. Similarly, the computational domain with PML is $[0, 12] \times [0, 12]$. The symmetric B.C. is given by $\partial p / \partial n = 0$, which is mathematically identical to the solid wall B.C. Therefore, the treatment is the same as the solid wall treatment.

To illustrate the validation and high accuracy of TSEM, three different methods including FEM (with triangular elements), SEM and TSEM are used. Three cases are tested, where in each case the total number of interpolation points Nd are kept as consistent for different methods. Parameter settings of the three cases are listed in Table 1. The element distributions for each method in Case 2 are showed in Figure 4. Theoretically, SEM and TSEM lead to more accurate numerical results whereas FEM introduces a lower accuracy with the same number of interpolation points. Figure 5 shows the pressure distribution for Case 2 on the line of $y = 0$ with x varies from 0 to 10 when $t = 2, 4$ and 6. Results of SEM and TSEM are very close to the analytical solutions, while FEM leads to obvious deviations, especially at the peak and valley of the pressure wave.

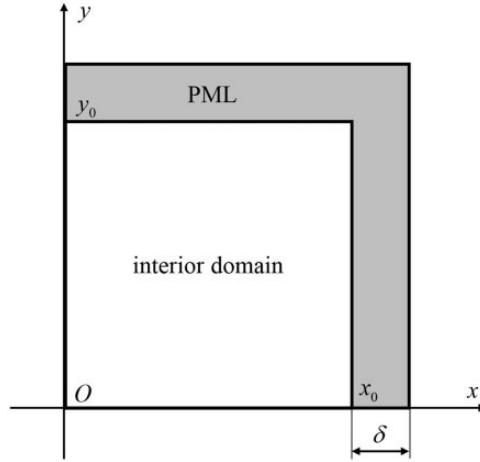


Figure 3. The computational domain of example I with PML region included.

Table 1. Parameter settings for accuracy test.

	FEM		SEM		TSEM		Nd
	NG	N	NG	N	NG	N	
Case 1	1250	/	25	5	50	5	676
Case 2	800	/	25	4	50	4	441
Case 3	512	/	16	4	32	4	289

The numerical accuracy is evaluated by L_2 and L_∞ norms of error:

$$L_2 = \sqrt{\sum_{i=1}^{Nd} \frac{(\tilde{p} - p)^2}{Nd}} \quad (54)$$

$$L_\infty = \max_{1 \leq i \leq Nd} |\tilde{p} - p| \quad (55)$$

where $\tilde{p}$ represents the numerical result, and p represents the analytical one. The L_2 norm for error and calculation time consumption of three methods are compared in Table 2. We calculate these cases using serial calculation in MATLAB with Intel(R) Core(TM) i7-7700K, 4.2 GHz. Time cost of FEM is about 1/3 smaller than SEM and TSEM, because matrices (**M**, **C** and **K**) for FEM is sparse due to its low order. However, the error made by FEM is nearly two orders of magnitude larger than SEM and TSEM, which meets the expectation.

A further discussion is made for TSEM to validate the convergence characteristic of error. Figure 6 shows the error made by TSEM under different parameter(Δt and N) settings. When Δt is small enough, errors decrease exponentially as N goes up, that is to say,

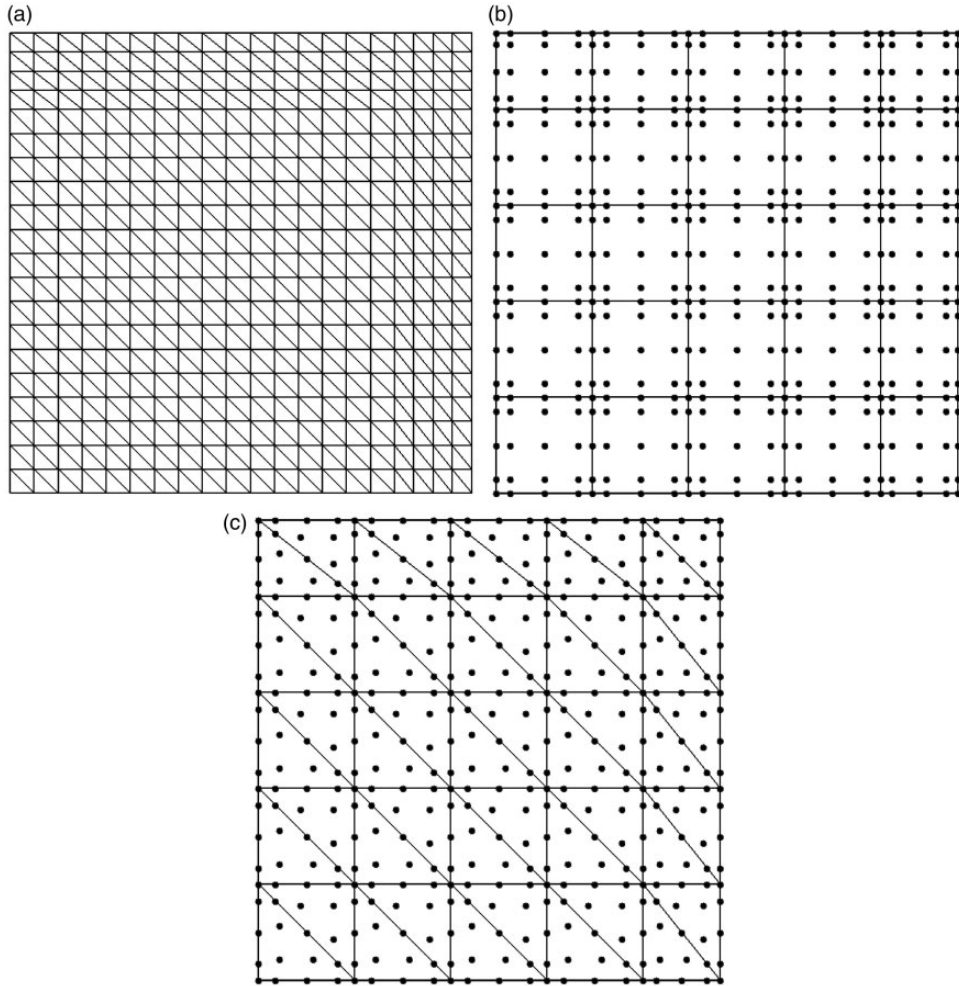


Figure 4. Distributions of elements and interpolation points for Case 2 (PML areas are included). (a) FEM, (b) SEM, (c) TSEM.

the exponential convergence rate can be reached. In other words, a simple operation of setting N to a larger value could increase numerical accuracy efficiently. This feature makes TSEM more convenient than high-order FDM.

Additionally, we investigate the effectiveness of PML by comparing PML with another widely used non-reflective B.C. proposed by Clayton, Engquist and Majda (denote by CEM).³⁷ First-order CEM is implemented by introducing a Neumann boundary condition on the artificial B.C.:

$$\frac{1}{c_\infty} \frac{\partial p}{\partial t} + \frac{\partial p}{\partial n} = 0 \quad (56)$$

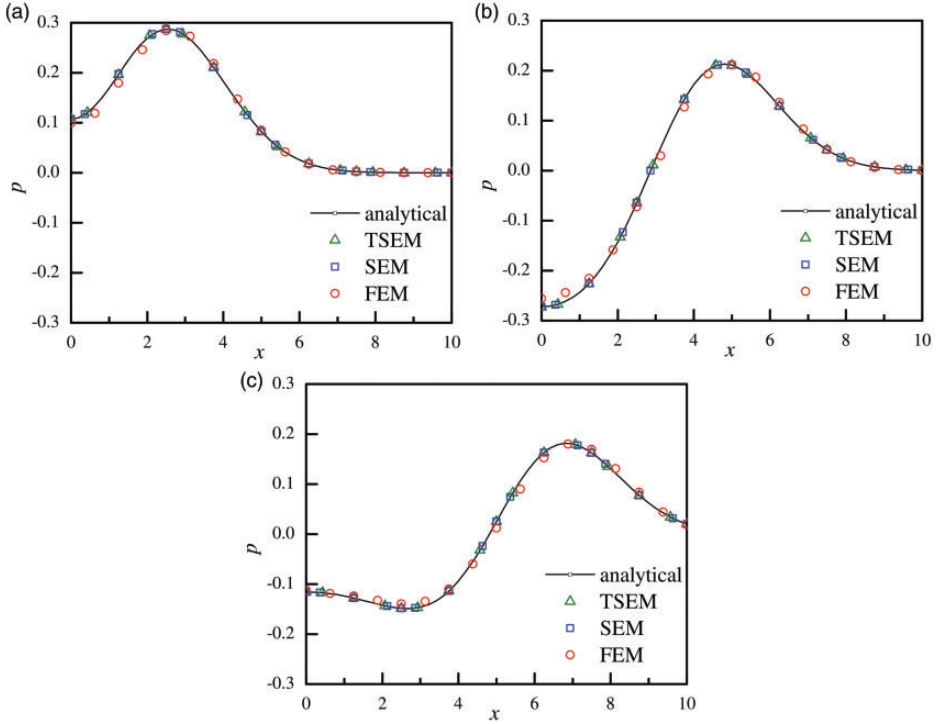


Figure 5. Pressure distributions for Case 2 at different time along $x = [0, 10]$ and $y = 0$. (a) $t = 2$, (b) $t = 4$, (c) $t = 6$.

Table 2. L_2 norm of error and time consumption of FEM, SEM and TSEM when $t = 6$.

	FEM		SEM		TSEM	
	$L_2 \times 10^5$	runtime(s)	$L_2 \times 10^5$	runtime(s)	$L_2 \times 10^5$	runtime(s)
Case 1	329.4	56.44	6.404	89.23	4.165	90.41
Case 2	507.8	30.86	9.754	40.06	14.12	44.49
Case 3	878.7	18.52	2.130	25.19	5.910	24.84

$\Delta t = 10^{-4}$, 60,000 cycles are run.

Figure 7 shows the pressure distribution at $t = 15$ when the wave propagates beyond the calculation domain to show the absorbing ability. Obvious wave reflection can be observed in CEM case, while PML with proper parameter selection leads to good agreement with analytical solution.

In Figure 7, we also discuss the effect of different absorbing parameters (σ and δ) on the accuracy. As shown in Figure 7(a), larger δ leads to better result, because large δ allows wave to be absorbed slowly and thoroughly, but also more time consuming. Figure 7(b) compares different σ . It seems effect of PML highly depends on choice of σ . When σ is too small, wave can't be absorbed completely. When σ is too large, the result becomes worse than CEM with huge derivation from the analytical solution observed,. A possible

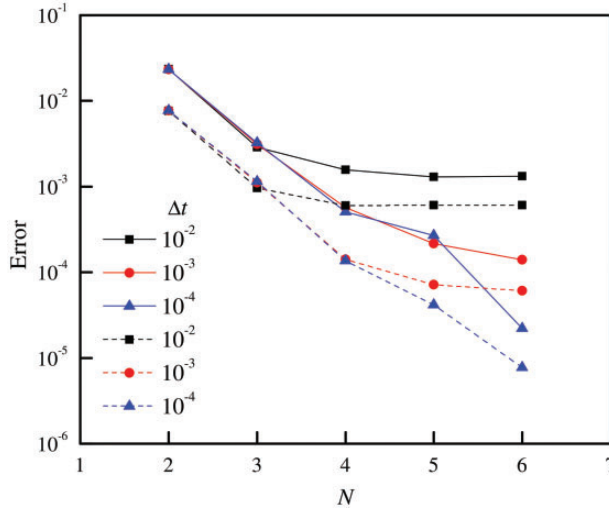


Figure 6. Convergence characteristic test: L_2 (dash line) and L_∞ (solid line) norms of error for pressure p when $t = 6$.

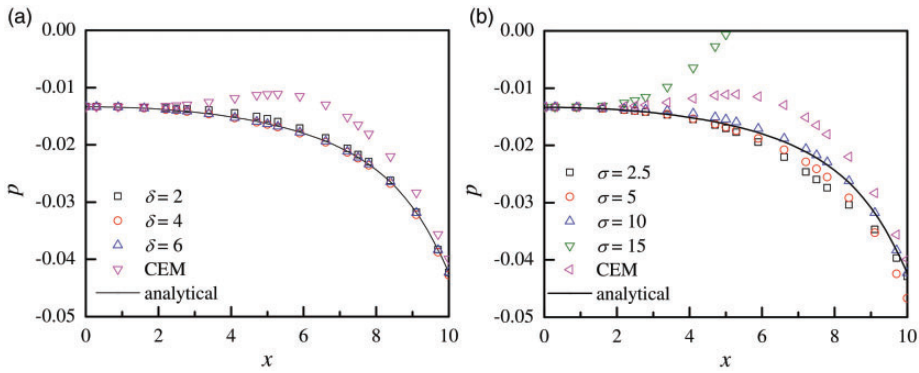


Figure 7. Test of absorbing parameters: pressure distributions at $t = 15$. (a) different δ with $\sigma = 10$, (b) different σ with $\delta = 2$.

explanation is that large pressure gradient is produced with large σ , and thus leads to spurious oscillation. A reference value given by Hu³ is $\sigma = 4/\Delta x$, where Δx is the distance between nodes in PML regions.

Sound reflection by a flat wall

The second example is a problem from ICASE/LaRC Workshop on Benchmark Problems in CAA.³⁸ As shown in Figure 8, an initial pulse located at $(0, 25)$ can be written as:

$$p(x, y, 0) = \exp \left[-\ln 2 \frac{x^2 + (y - 25)^2}{25} \right] \quad (57)$$

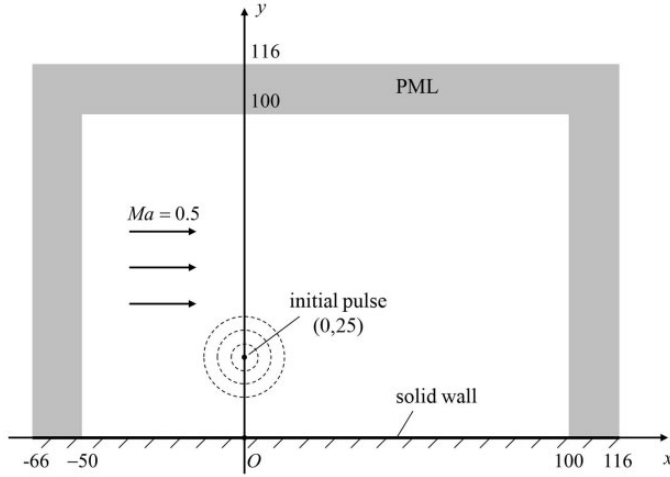


Figure 8. Schematic of example 2: wave propagation and reflection in uniform mean flow.

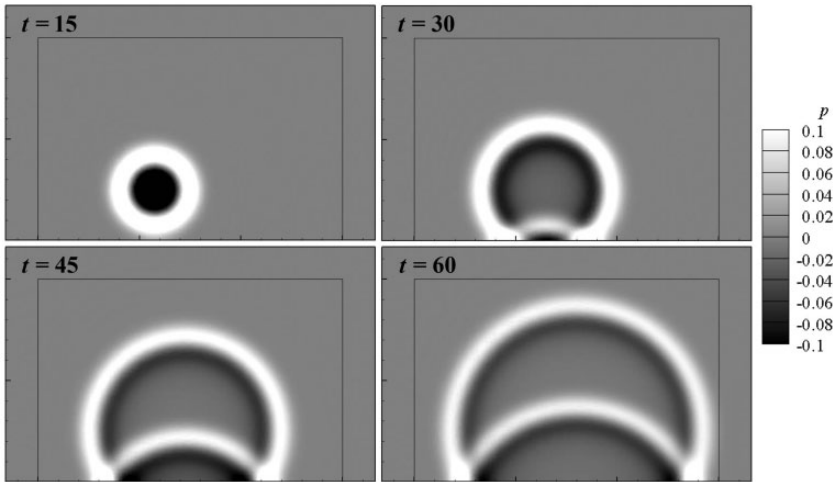


Figure 9. Pressure contours of sound reflection from a flat wall, at $t = 15, 30, 45$ and 60 . Regions outside the rectangle are PML regions.

A flat solid wall is placed at $y = 0$. The mean flow is along x -axis with $Ma = 0.5$. In this benchmark problem, we aim to testify the validation of solid wall B.C. and the calculation when $Ma \neq 0$. The calculation is performed in a region of $[-50, 100] \times [0, 100]$, the PML is set as $\delta = 16$. We set $N = 5$, totally 616 elements and 7881 interpolation points are used. The time increment is 0.005.

In Figure 9, pressure distributions at $t = 15, 30, 45$ and 60 are given. Reflected wave from the flat wall is clearly observed. In Figure 10, computation results are compared with analytical solutions along the line $y = 25$. A good agreement between analytical solutions and

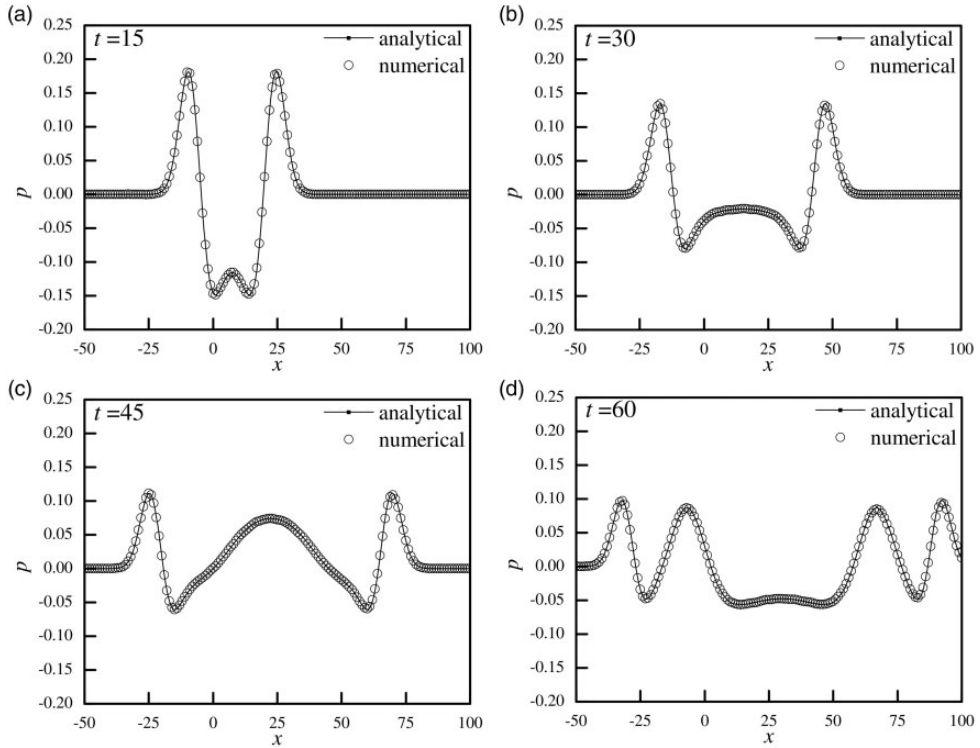


Figure 10. Pressure distributions at different time along $y=25$ and $x = [-50, 100]$.

results by this present method can be seen, which suggests the solid wall B.C. is successfully implemented.

Sound scattering around a circular cylinder

Example 3 is a benchmark problem from Second CAA Workshop on Benchmark Problems.³⁹ Here it's used to test the ability to describe complex geometries. As shown in Figure 11, there's an initial pulse centered on the origin:

$$p(x, y, 0) = \exp\left(-\ln 2 \frac{x^2 + y^2}{0.04}\right) \quad (58)$$

A circular cylinder with $r=0.5$ is located at $(-4, 0)$. The surface of the cylinder is assumed to be rigid wall, so the solid wall B.C. is applied on the wall. Due to the symmetry about x axis, only upper half of the domain is calculated. Hence the calculation is made over a domain of $[-9, 5] \times [0, 5]$ with PML included. PML is used for outer artificial boundaries with a thickness of $\delta=2$. The mesh showed in Figure 11 is generated by DistMesh.⁴⁰ The total number of triangular elements is $NG=735$ (PML included). After grid refinement

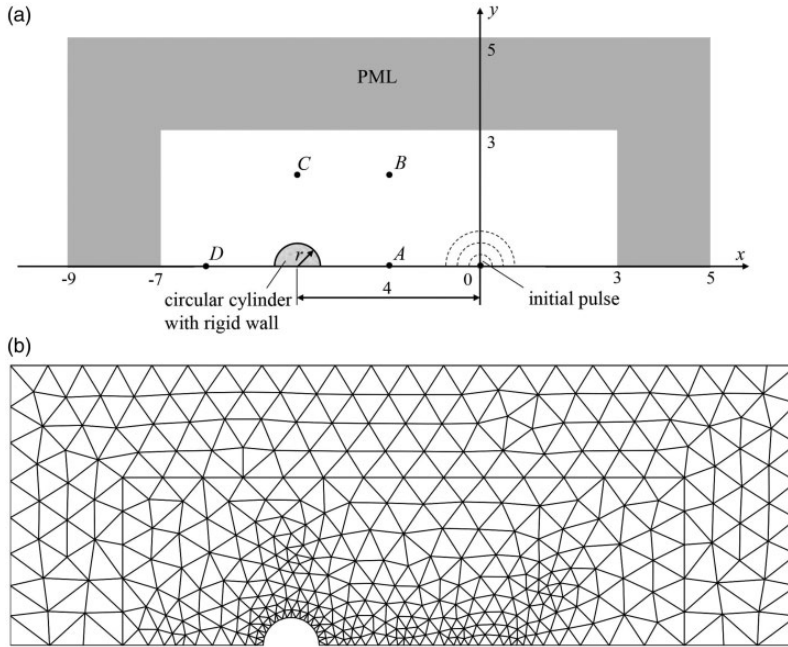


Figure 11. Sound wave generated by initial pulse scatters around a circular cylinder with $r=0.5$. (a) Schematic diagram; (b) Mesh distribution.

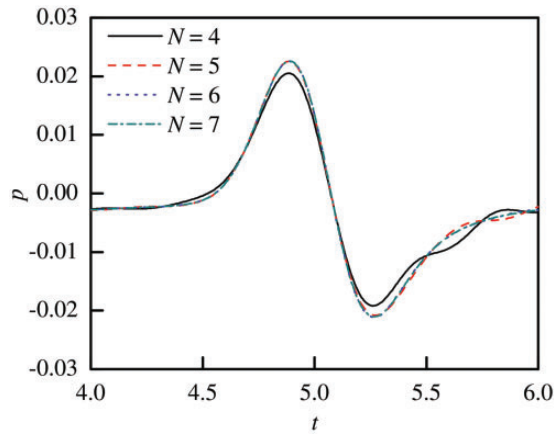


Figure 12. Grid refinement study for example 3: time history of pressure at point $A(-2, 0)$.

analysis (see Figure 12), $N=6$ is chosen as the interpolation order, and the total number of interpolation points is $Nd=13,732$. As showed in Figure 11, 20 triangle elements are used to approximate the semicircular rigid wall (40 elements for the whole circle) to describe the shape precisely.

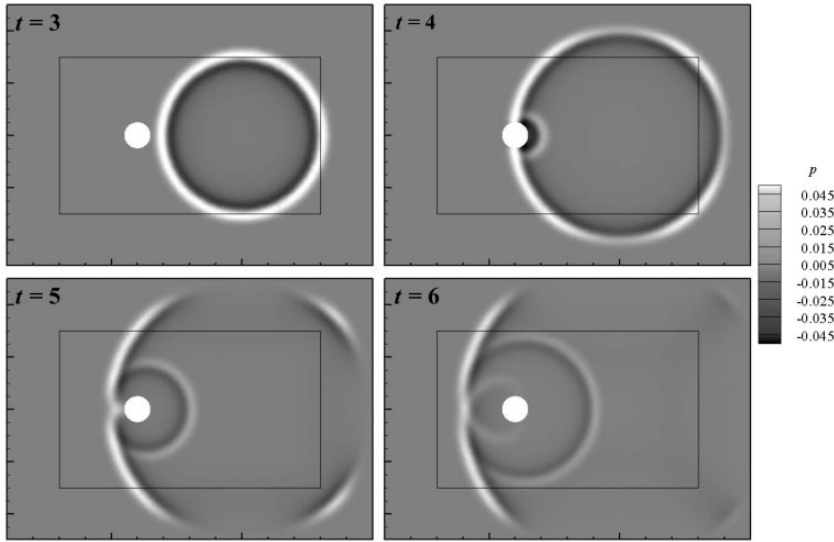


Figure 13. Pressure distribution of sound scattering around a circular cylinder at $t = 3, 4, 5$ and 6 . Area outside the rectangle is the PML region. The lower half is the mirror image about the x -axis.

Pressure distributions at $t = 3, 4, 5$ and 6 are showed in Figure 13. These pictures depict how the incident wave generated by the initial pulse propagates, and how the wave reflects under the effect of solid wall. Figure 14 shows the time histories of p monitored at four points including $A(-2, 0)$, $B(-2, 2)$, $C(-4, 2)$ and $D(-6, 0)$. Compared with the analytical solution, the reflected wave is accurately captured by TSEM calculation.

Helmholtz resonator

Resonance frequency of HR. For the last example, TSEM is used to solve the pressure field in Helmholtz resonators (HRs). A typical 2D HR consists of a neck and a large resonant cavity (see Figure 15). When a disturbance is given outside the orifice of the neck, wave near a specific frequency f_0 will be enhanced.⁴¹ f_0 is known as the resonance frequency (RF) of HR, when the volume of the neck is negligible in comparison with the cavity, a theoretical approximation of f_0 is given as:

$$f_0 = \frac{c_\infty}{2\pi} \sqrt{\frac{b}{l\pi R^2}} \quad (59)$$

where b and l are the width and length of the neck, R is the radius of the cavity as showed in Figure 15, c_∞ is speed of sound. When the size of the neck is considerable, an approximate correction is added to l : $l' = l + 0.8b$. Sound near f_0 could be absorbed by HRs effectively. Take advantage of this feature, HRs are applied to noise reduction, especially low-frequency noise. There are some researches exploring possibilities of applying HRs on gas turbine combustors,⁴² tunnels for high-speed trains,⁴³ rocket fairings⁴⁴ and so on.

This example is aimed to verify the resonance characteristic and the performance of HRs to reduce noise. The purpose is to show the possibility of dealing with practical problems

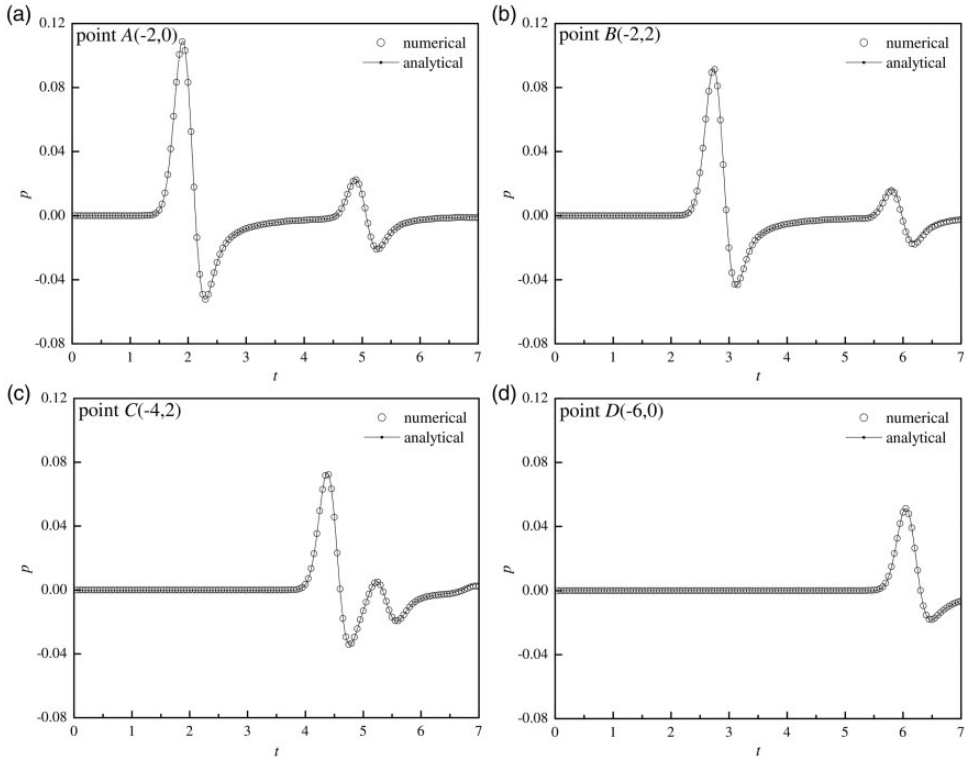


Figure 14. Time histories of pressure monitored at four different points.

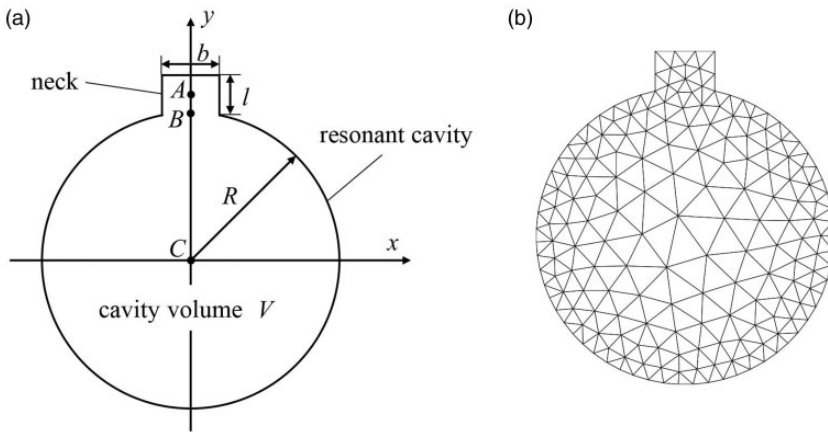


Figure 15. A typical Helmholtz resonator with a neck of width b and length l and a resonant cavity of volume V and radius R .

(a) Schematic diagram, (b) Distribution of triangular elements.

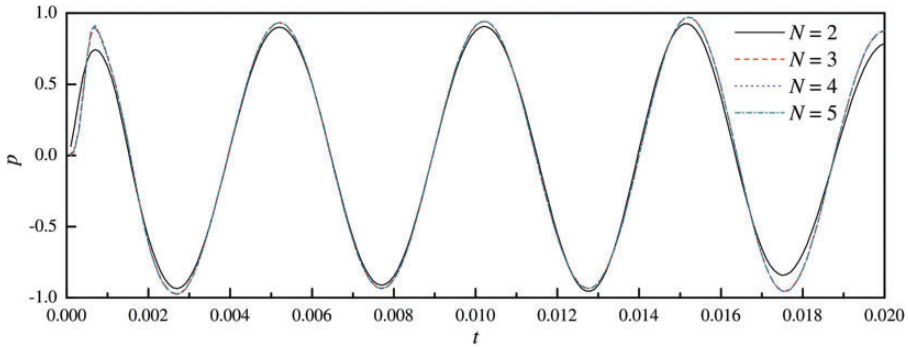


Figure 16. Grid refinement analysis for HR: time history of pressure at point $A(0, 1.125)$ when $f = 200$ Hz.

using TSEM. We set $b = 0.4$ m, $l = 0.25$ m, $R = 1$ m, $c_\infty = 340$ m/s, so the theoretical estimation of f_0 is 25.57 Hz. To verify f_0 numerically, we set B.C. of the first kind at the orifice of the neck:

$$p(t) = \cos(2\pi ft) \quad (60)$$

where f ranges from 10 Hz to 200 Hz. A grid refinement analysis is carried out in Figure 16 for the highest frequency 200 Hz. The interpolation order is set as $N = 4$, totally 316 elements and 2661 points are used. Time increment is 0.0001 s, and 10,000 steps are calculated for each f . Pressure is monitored at three points including $A(0, 1.125)$, $B(0, 1)$, and $C(0, 0)$.

Figure 17 depicts the pressure change with time when f is 23 Hz and 25 Hz. The amplitude of p changes periodically at $f = 23$ Hz, while the amplitude grows larger and larger at $f = 25$ Hz. Pressure differs significantly though the two frequencies are very close, which may suggest the existence of RF near them.

We introduce an "average power" of p to evaluate the response of HR when f varies:

$$\bar{P} = \frac{1}{T} \int_0^T p^2 dt = \frac{\Delta t}{T} \sum_{i=1}^{T/\Delta t} p_i^2 \quad (61)$$

where p_i is the pressure when $t = i \times \Delta t$.

Frequencies near the RF will lead to high response from the HR and high value of $\bar{P}$. Figure 18 illustrates how $\bar{P}$ changes when f varies from 10 Hz to 200 Hz. The first and highest peak is near 25 Hz, which is close to the theoretical value of 25.57 Hz. Additionally, there are other two lower peaks near 109 and 170 Hz, which could be regarded as the second and third order of RFs. It's also noticed from Figure 18 that for point C (inside the HR), $\bar{P}$ at 25 Hz is much higher than that at the other two RFs, while $\bar{P}$ distributes more evenly for point A and B (close to the orifice).

HR connected with a tube. A monopole source is assumed to be in a straight tube with infinite length (see Figure 19(a)) to simulate noises. Then we attach a HR to the tube to absorb

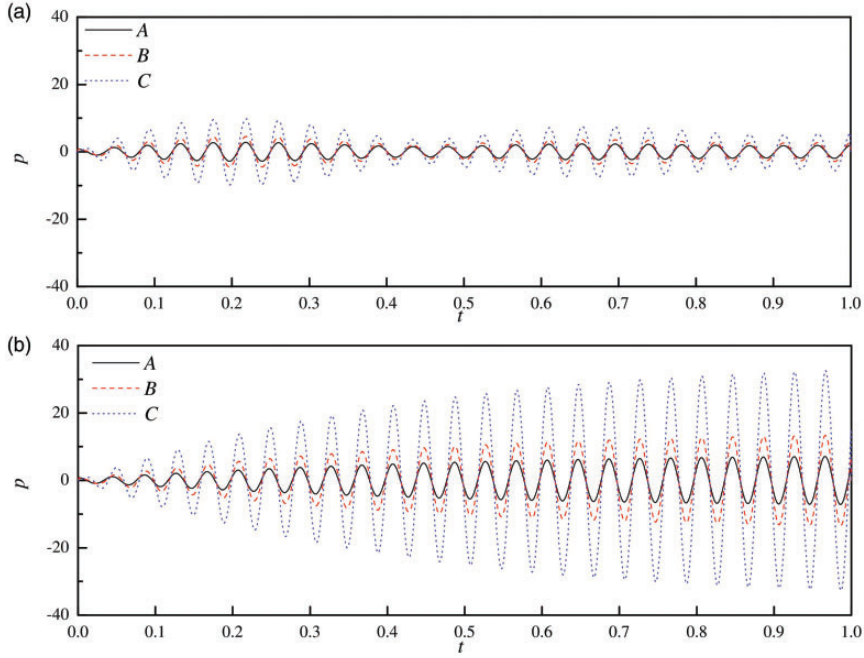


Figure 17. Pressure change at point $A(0, 1.125)$, $B(0, 1)$, $C(0, 0)$.

(a) $f = 23$ Hz, (b) $f = 25$ Hz.

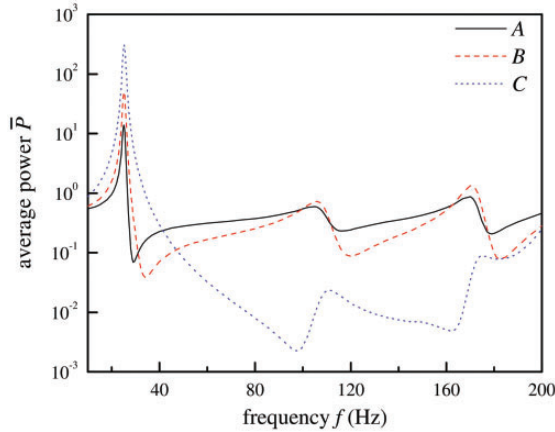


Figure 18. Response curve of Helmholtz resonator: average power of p at point A , B and C .

them. The HR is exactly the same size as given above. A section of tube with length $L = 8$ near the HR is selected as the computational domain. The width of the tube D is 2. The monopole S is located at $(-3.5, 0)$, with expression $S = \exp\{-\ln 2[(x + 3.5)^2 + y^2]/0.16\} \cos(2\pi f t)$. The frequency f ranges from 10 Hz to 200 Hz. To test the absorption

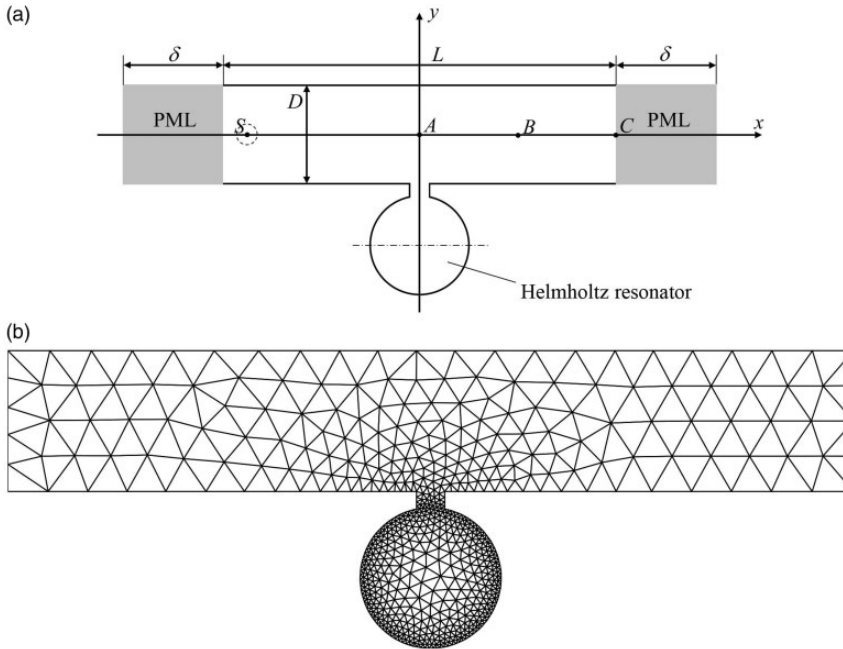


Figure 19. An infinite tube attached to a Helmholtz resonator.
(a) Schematic diagram, (b) Distribution of elements.

effect, pressure distribution in the tube is computed both with and without the HR. Three points $A(0, 0)$, $B(2, 0)$ and $C(4, 0)$ are monitored (see Figure 19(a)). Element distribution is given in Figure 19(b). The interpolation order is the same as calculation for HR, $N=4$, totally 1171 elements and 9739 points are used.

There are generally three typical modes of pressure distributions when f varies. Figure 20 shows pressure contours of the three modes, which are represented by $f=20\text{Hz}$, 100Hz and 180Hz . Pressure changes periodically in the same frequency as the monopole when time is long enough (usually much longer than the period of the monopole). When the frequency is extremely low, p changes slowly along the x -axis while the value keeps almost the same in the cross section. When f goes up to about 100Hz , rapid pressure change along the x -axis and inside the HR appears. When f reaches about 180Hz , the mode with significant pressure changes in both the cross section and x -axis is observed, and amplitude of pressure on the right side of the tube is much smaller than amplitude on the left side.

$\bar{P}$ without HR attached when f varies is compared in Figure 21(a). When f is low, $\bar{P}$ at each point is almost the same with only tiny differences. $\bar{P}$ decreases with a nearly exponential rate as f increases. Faster descent can be observed for point B and C than A . An interesting phenomenon is that when f grows to 160Hz , value of $\bar{P}$ begin to go up, and difference between curves of $\bar{P}$ at three points emerges. For point A and B , $\bar{P}$ keeps increasing till f reaches about 180Hz , then $\bar{P}$ becomes lower. The width of the peak is larger for point A than point B . For point C , change of $\bar{P}$ is an S-shape curve when f ranges from 160Hz to 200Hz , which is totally different from the other two points. This finding is in agreement with three different modes of pressure distributions we mentioned before.

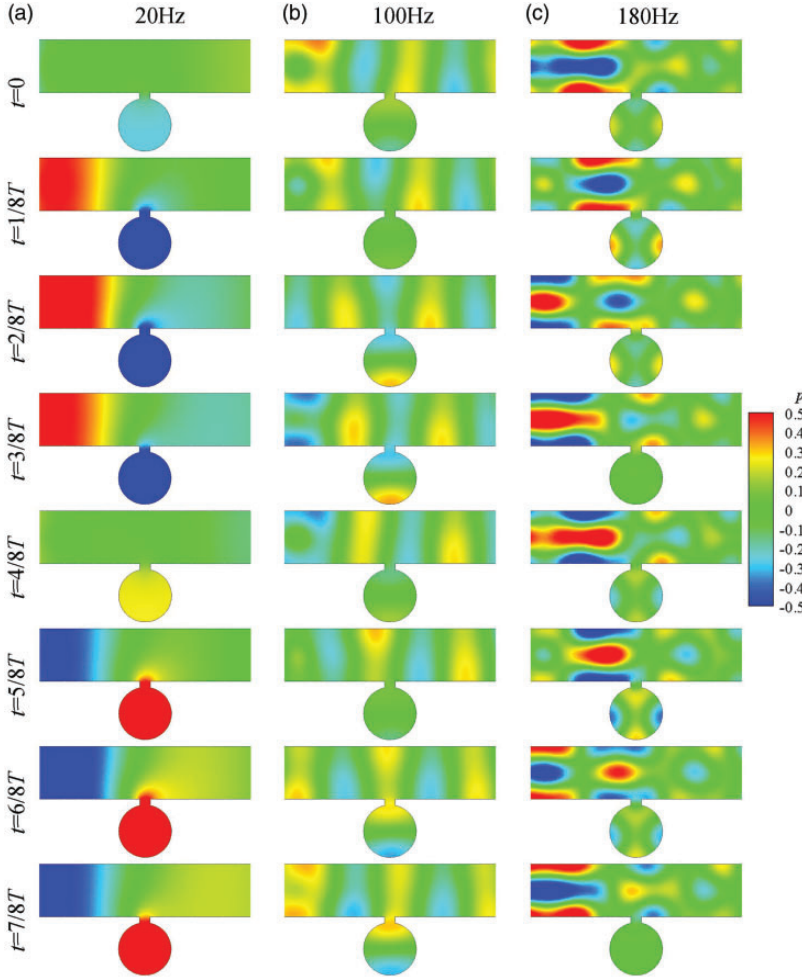


Figure 20. Three typical modes of pressure distribution in the tube with HR connected when frequency varies. One period is shown for each frequency, $T = 1/f$. $t=0$ is set as a moment when steady periodic condition is reached. Pressure values of $f = 100\text{Hz}$ and 180Hz are tripled for visual exhibition.

When f is low, $\bar{P}$ distributes evenly in the tube due to a uniformly distribution of p with a time integration. When f is large enough to transfer into the third mode (see Figure 20(c)), $\bar{P}$ on the left side is significantly larger than the right, which leads to diversity of $\bar{P}$ at different points.

Considering effect of the HR, as shown in Figure 21(b), it's obvious that HR does make $\bar{P}$ change significantly, especially when f is close to RFs. To evaluate the absorption of HR quantitatively, we introduce the absorptivity ε :

$$\varepsilon = \bar{P}' / \bar{P}_0 \quad (62)$$

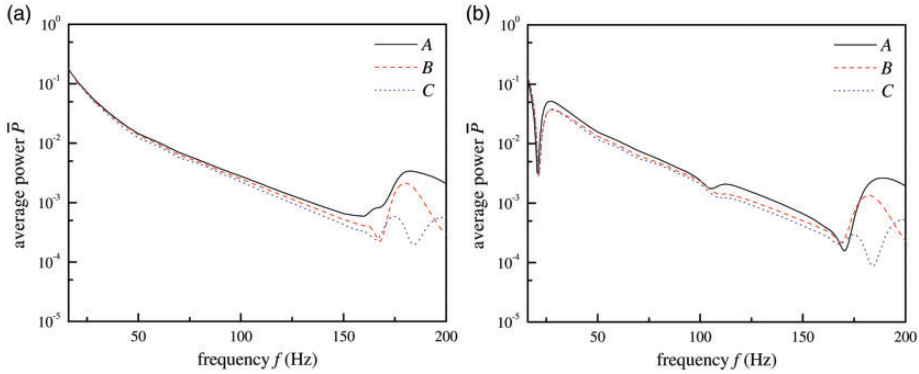


Figure 21. $\bar{P}$ at point A(0, 0), B(2, 0) and C(4, 0), depending on the frequency of monopole. (a) Without HR attached, (b) With HR attached.

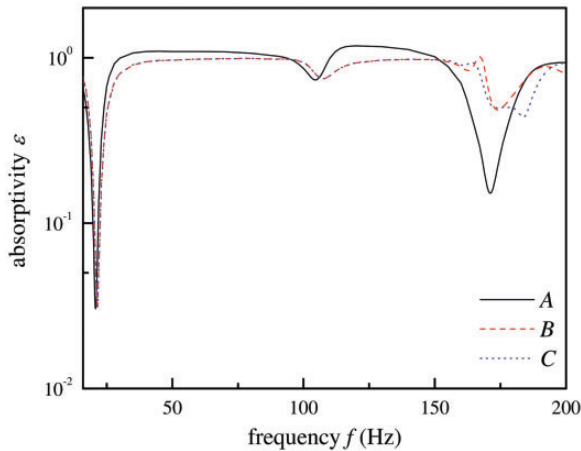


Figure 22. Absorptivity of pressure at different frequencies at three points.

where the subscript ‘0’ denotes tube without HR and the superscript ‘ $\prime$ ’ represents tube with HR. The smaller ε is, the better the noise reduction is. Figure 22 shows the absorptivity at all three points. ε is less than 1 at nearly every frequency, therefore, the effect of HR to reduce noise is validated. Three valley values, where noises are absorbed with high efficiency, are corresponding to three RFs. However, because of narrow peaks on the response curve of HR (Figure 18), these valleys are narrow too, i.e. the band where noises could be absorbed well is narrow. Noise in frequencies located between RFs can’t be reduced efficiently. It should be noted that, all the viscosity and energy losses are ignored in the present paper. If take losses into consideration, the absorption band will be wider, which means the capability of HRs is underestimated in this paper. Additionally, to increase the bandwidth and improve the capability of absorption to a higher degree, changing the geometry of the neck,⁴⁵ combining with porous materials⁴⁶ or using microperforated panels (distributed HRs)⁴⁷ may be feasible.

Another interesting finding is that, the valleys for first and second order of RFs appear at around $f = 21\text{Hz}$ and $f = 104 - 108\text{Hz}$ (the value is different at each point). Generally speaking, the resonance points shift to lower value compared with RFs in the original HR. A possible explanation is that the attached tube can be considered as an extension of the neck of HR, so in equation (59), an extra correction should be added to the length of neck l . In this way, the RF becomes smaller than the situation without the tube.

Conclusion

In this paper, an accurate triangular spectral element method is presented on CAA problems with complex domains. The implementation of triangular elements improves the flexibility of discretizing complex geometry. The second-order governing equations instead of the original linearized Euler equations contribute to both calculation parameter reducing and solid wall boundary conditions handling. Through comparison with analytical solution, FEM and SEM, the high accuracy (compared with FEM) and exponential convergence of TSEM are validated. Three benchmark problems show the effectiveness of this present method when dealing with acoustic problems in complex geometries. The application on Helmholtz resonator calculation demonstrates the noise reduction function of HRs, which reaches an excellent agreement with theoretical estimation. The present TSEM provides a new solution for challenging CAA problems, and our future research will focus on applying TSEM to more complicated practical aeroacoustic problems in complex geometries.

Declaration of conflicting interests

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Supplemental material

Supplemental material for this article is available online.

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Appendix

Notation

p	dimensionless pressure
$\dot{p}, \ddot{p}$	first and second order derivatives of time for p
p_1, p_2	components of p

u, v	components of dimensionless velocity
ρ	dimensionless density
t	dimensionless time
c_∞	speed of sound(m/s)
ρ_∞	density of the uniform mean flow(kg/m ³)
Ma	Mach number of the mean flow
Ma_x, Ma_y	components of Ma
S	source term
σ_x, σ_y	absorption coefficients of PML
Ω	standard triangle element
$\Phi_i(x, y)$	i th Koornwinder-Dubiner polynomial
$P_i^{\alpha, \beta}(x)$	Jacobi polynomial in i th order
N	order of Fekete points and KD polynomials
N_p	number of Fekete points in a triangular element
NG	number of triangular elements
Nd	total number of Fekete points in the computational domain
$\mathbf{V}$	Vandermonde matrix
L	Lagrange basis
v	test function
$h^{(i)}$	value of h at i th interpolation point, h can be any scalar
f_0	resonance frequency of Helmholtz resonator
$\bar{P}$	“average power” of pressure
ε	absorption coefficient of Helmholtz resonator

Appendix I. Derivation of governing equations about p

The linearized Euler equations given in equation (2) can be rewritten as:

$$\frac{\partial \rho}{\partial t} + Ma_x \frac{\partial \rho}{\partial x} + \frac{\partial u}{\partial x} + Ma_y \frac{\partial \rho}{\partial y} + \frac{\partial v}{\partial y} = S_1 \quad (\text{B1})$$

$$\frac{\partial u}{\partial t} + Ma_x \frac{\partial u}{\partial x} + \frac{\partial p}{\partial x} + Ma_y \frac{\partial u}{\partial y} = S_2 \quad (\text{B2})$$

$$\frac{\partial v}{\partial t} + Ma_x \frac{\partial v}{\partial x} + Ma_y \frac{\partial v}{\partial y} + \frac{\partial p}{\partial y} = S_3 \quad (\text{B3})$$

$$\frac{\partial p}{\partial t} + Ma_x \frac{\partial p}{\partial x} + \frac{\partial u}{\partial x} + Ma_y \frac{\partial p}{\partial y} + \frac{\partial v}{\partial y} = S_4 \quad (\text{B4})$$

The partial derivative of equation (B2) respect to x is:

$$\frac{\partial^2 u}{\partial t \partial x} + Ma_x \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 p}{\partial x^2} + Ma_y \frac{\partial^2 u}{\partial x \partial y} = \frac{S_2}{\partial x} \quad (\text{B5})$$

Similarly, the partial derivative of equation (B3) respect to y is:

$$\frac{\partial^2 v}{\partial t \partial y} + Ma_x \frac{\partial^2 v}{\partial x \partial y} + Ma_y \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 p}{\partial y^2} = \frac{S_3}{\partial y} \quad (B6)$$

The partial derivatives of equation (B4) respect to x , y and t can be expressed by:

$$\frac{\partial^2 p}{\partial t \partial x} + Ma_x \frac{\partial^2 p}{\partial x^2} + \frac{\partial^2 u}{\partial x^2} + Ma_y \frac{\partial^2 p}{\partial x \partial y} + \frac{\partial^2 v}{\partial x \partial y} = \frac{\partial S_4}{\partial x} \quad (B7)$$

$$\frac{\partial^2 p}{\partial t \partial y} + Ma_x \frac{\partial^2 p}{\partial x \partial y} + \frac{\partial^2 u}{\partial x \partial y} + Ma_y \frac{\partial^2 p}{\partial y^2} + \frac{\partial^2 v}{\partial y^2} = \frac{\partial S_4}{\partial y} \quad (B8)$$

$$\frac{\partial^2 p}{\partial t^2} + Ma_x \frac{\partial^2 p}{\partial t \partial x} + \frac{\partial^2 u}{\partial t \partial x} + Ma_y \frac{\partial^2 p}{\partial t \partial y} + \frac{\partial^2 v}{\partial t \partial y} = \frac{\partial S_4}{\partial t} \quad (B9)$$

$Ma \times \text{equation (B7)} + Ma \times \text{equation (B8)} + \text{equation (B9)}$ and substitute equations (B5)-(B6) in, we get:

$$\frac{\partial^2 p}{\partial t^2} + 2Ma_x \frac{\partial^2 p}{\partial t \partial x} + 2Ma_y \frac{\partial^2 p}{\partial t \partial y} + 2Ma_x Ma_y \frac{\partial^2 p}{\partial x \partial y} - \left[(Ma_x^2 - 1) \frac{\partial^2 p}{\partial x^2} + (Ma_y^2 - 1) \frac{\partial^2 p}{\partial y^2} \right] = S \quad (B10)$$

$$S = \frac{\partial S_2}{\partial x} + \frac{\partial S_3}{\partial y} + Ma_x \frac{\partial S_4}{\partial x} + Ma_y \frac{\partial S_4}{\partial y} + \frac{\partial S_4}{\partial t} \quad (B11)$$

Appendix 2. Derivation of PML-included governing equations

The governing equations without PML are:

$$\frac{\partial^2 p}{\partial t^2} + 2Ma \frac{\partial^2 p}{\partial t \partial x} + \left[(Ma^2 - 1) \frac{\partial^2 p}{\partial x^2} - \frac{\partial^2 p}{\partial y^2} \right] = S \quad (C1)$$

which could be rewritten into the frequency domain:

$$(i\omega)^2 \tilde{p} + 2Ma i\omega \frac{\partial \tilde{p}}{\partial x} + \left[(Ma^2 - 1) \frac{\partial^2 \tilde{p}}{\partial x^2} - \frac{\partial^2 \tilde{p}}{\partial y^2} \right] = \tilde{S} \quad (C2)$$

where the hat ‘ $\sim$ ’ denotes corresponding quantity of p in the frequency domain.

The ordinate transformations of PML are:

$$\frac{\partial}{\partial x} = \frac{i\omega}{i\omega + \sigma_x} \frac{\partial}{\partial x'} \quad (C3)$$

$$\frac{\partial}{\partial y} = \frac{i\omega}{i\omega + \sigma_y} \frac{\partial}{\partial y'} \quad (C4)$$

Substitute equations (C3)-(C4) into equation (C2):

$$(i\omega)^2 \tilde{p} + 2Ma \frac{(i\omega)^2}{i\omega + \sigma_x} \frac{\partial \tilde{p}}{\partial x'} + (Ma^2 - 1) \frac{i\omega}{i\omega + \sigma_x} \frac{\partial}{\partial x'} \left(\frac{i\omega}{i\omega + \sigma_x} \frac{\partial \tilde{p}}{\partial x'} \right) - \frac{i\omega}{i\omega + \sigma_y} \frac{\partial}{\partial y'} \left(\frac{i\omega}{i\omega + \sigma_y} \frac{\partial \tilde{p}}{\partial y'} \right) = \tilde{S} \quad (C5)$$

In following derivations, the subscript ‘’ is omitted for the sake of convenience. We split equation (C5) into two equations:

$$(i\omega)^2 \tilde{p}_1 + 2Ma \frac{(i\omega)^2}{i\omega + \sigma_x} \frac{\partial \tilde{p}}{\partial x} + (Ma^2 - 1) \frac{i\omega}{i\omega + \sigma_x} \frac{\partial}{\partial x} \left(\frac{i\omega}{i\omega + \sigma_x} \frac{\partial \tilde{p}}{\partial x} \right) = \frac{\tilde{S}}{2} \quad (C6)$$

$$(i\omega)^2 \tilde{p}_2 - \frac{i\omega}{i\omega + \sigma_y} \frac{\partial}{\partial y} \left(\frac{i\omega}{i\omega + \sigma_y} \frac{\partial \tilde{p}}{\partial y} \right) = \frac{\tilde{S}}{2} \quad (C7)$$

where $\tilde{p}_1$ and $\tilde{p}_2$ are components of $\tilde{p}$, $\tilde{p} = \tilde{p}_1 + \tilde{p}_2$. Their corresponding quantities in the time domain have the same relationship: $p = p_1 + p_2$. Here the source term S in equation (C5) is split into $S/2$ and $S/2$ in equations (C6)-(C7) for demonstration. But actually any other split that guarantees the sum of the two split source terms equal 1, say, $3S/4$ and $S/4$ is feasible.

Equation (C6) $\times [(i\omega + \sigma_x)/i\omega]^2$, we have:

$$(i\omega + \sigma_x)^2 \tilde{p}_1 + 2Ma(i\omega + \sigma_x) \frac{\partial \tilde{p}}{\partial x} + (Ma^2 - 1) \left(\frac{\partial^2 \tilde{p}}{\partial x^2} + \frac{1}{i\omega + \sigma_x} \frac{\partial \tilde{p}}{\partial x} \frac{\partial \sigma_x}{\partial x} \right) = \frac{(i\omega + \sigma_x)^2}{(i\omega)^2} \frac{\tilde{S}}{2} \quad (C8)$$

Assume all the sources are in the interior region where $\sigma_x = 0$, σ_x on the right hand side of equation (C8) can be omitted.

Here we introduce an intermediate parameter $\tilde{f}$:

$$\tilde{f} = \frac{1}{i\omega + \sigma_x} \frac{\partial \tilde{p}}{\partial x} \frac{\partial \sigma_x}{\partial x} \quad (C9)$$

Substitute equation (C9) into equation (C8):

$$(i\omega + \sigma_x)^2 \tilde{p}_1 + 2Ma(i\omega + \sigma_x) \frac{\partial \tilde{p}}{\partial x} + (Ma^2 - 1) \left(\frac{\partial^2 \tilde{p}}{\partial x^2} + \tilde{f} \right) = \frac{(i\omega + \sigma_x)^2}{(i\omega)^2} \frac{\tilde{S}}{2} \quad (C10)$$

Rewrite equation (C10) and equation (C9) into the time domain:

$$\left(\frac{\partial}{\partial t} + \sigma_x \right)^2 p_1 + 2Ma \left(\frac{\partial}{\partial t} + \sigma_x \right) \frac{\partial p}{\partial x} + (Ma^2 - 1) \left(\frac{\partial^2 p}{\partial x^2} + f \right) = \frac{S}{2} \quad (C11)$$

$$\left(\frac{\partial}{\partial t} + \sigma_x\right)f = \frac{\partial p}{\partial x} \frac{\partial \sigma_x}{\partial x} \quad (\text{C12})$$

Similarly, governing equation for p_2 can be obtained from equation (C7):

$$\left(\frac{\partial}{\partial t} + \sigma_y\right)^2 p_2 - \frac{\partial^2 p}{\partial y^2} - g = \frac{S}{2} \quad (\text{C13})$$

$$\left(\frac{\partial}{\partial t} + \sigma_y\right)g = \frac{\partial p}{\partial y} \frac{\partial \sigma_y}{\partial y} \quad (\text{C14})$$

Equations (C11)-(C14) are the PML-included governing equations.